

Rock Evidence Lab: Look Before You Label

I can compare rock specimens using observed properties and create a group using a stated rule.

OBSERVE

names stay hidden

COMPARE

same features

GROUP

use evidence

[Open the original interactive lesson](#)

What success looks like

- I can observe at least two properties.
- I can compare two coded specimens.
- I can state and apply one grouping rule.

Retrieve and reconnect

- Name the label basis kept the same in W8A.
- Define property as something we can observe or test.
- Explain why a sorting rule must be stated.
- Lead-in: give one observation without naming the rock.

Two rocks are both grey. Is that enough evidence to place them in the same useful group?

Think first. Point, say, sign or write your choice. Give one reason.

A. Yes. One shared colour is all we need for every purpose.

B. Not yet. Observe the same extra features on both rocks, then state a grouping rule.

C. Guess both names first and make the evidence fit later.

Look closely · what is the evidence?

CODE A · speckled surface with visible interlocking crystals

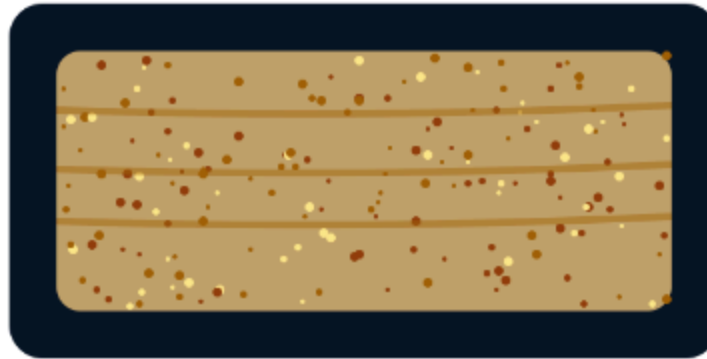
CODE C · rough surface with sand-sized grains and faint bands

QUESTION · which observed feature will become the grouping rule?

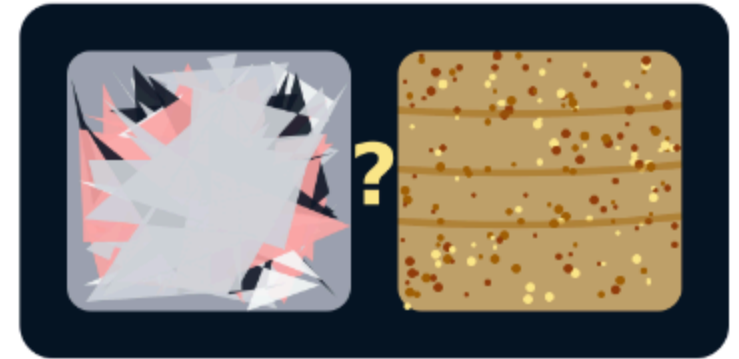
Look closely · what is the evidence?



Coded rock A: speckled with crystals



Coded rock C: sandy grains with faint bands



Rock A beside rock C with a question mark

Words we will use

- property — a feature we can notice or test
- observe — look closely and record
- classify — put into groups with a rule

Watch the model

- Keep the code and hide the name. Place one specimen on a white tray and observe it from more than one angle.
- Select two or three useful features: grains, crystals, bands, sheet-like planes, holes, colour or surface appearance.
- Record only what is visible. 'It is granite' is a name claim; 'several crystals are visible' is an observation.

Watch the model



Magnified view of the chosen coded rock

Which line is an observation that can be checked on the supplied specimen?

A. Several large crystals are visible in the speckled surface.

B. It must be granite because that is my first guess.

C. It will make the best roof because it looks strong.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Inspect a second coded specimen using the same feature words. State one similarity and one difference.
- Choose one rule, such as 'visible bands or flat sheet-like planes', and say the rule before grouping.
- Apply the same meaning to every specimen. Another clear rule may create a different valid grouping.

Find and repair the misconception

Rock glitch: if two rocks have the same colour, they must be the same type.

A. Keep it: colour always proves identity.

B. Repair it: compare several observed properties and keep identity claims cautious.

C. Repair it: group them by which colour the class likes most.

Coded Rock Feature Lab

Place each coded observation under its strongest named visible feature. The name is not the target.

- Choose a coded observation.
- Read only the visible evidence.
- Commit to the strongest feature group.
- Repair any mismatch.
- Use one feature as a grouping rule.

Coded Rock Feature Lab

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- VISIBLE CRYSTALS
- VISIBLE GRAINS
- BANDS / FLAT PLANES
- LARGE HOLES
- FINE · NO LARGE FEATURE

Choose one visible feature as a rule. Which codes belong in the yes group, and which evidence supports each placement?

Your independent mission

Inspect or direct the inspection of coded rock specimens, compare two using the same features and create two groups with a stated rule.

Record: A Rock Comparison Ticket with two codes, two observed properties, one similarity, one difference and one explicit grouping rule.

Choose your support and challenge

- Supported: Compare two coded macro cards with a four-word feature bank, then place them using one shown rule.
- Standard: Observe four to six coded specimens, compare a pair and create two groups using one stated property rule.
- Stretch: Create two different valid groupings, explain why both work and state why neither proves every specimen's identity.

Show what you know

Compare two coded specimens and defend one group using observed properties.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.